

ASSIGNMENT 3

QUESTION – 1 - Sum of Two Numbers (Using a Method)

```
import java.util.Scanner;

class sumOfTwoNumbers
{
    static int sumOfTwoNumbers(int a, int b) {
        int sum = a + b;
        return sum;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter 1st Number = ");
        int num1 = sc.nextInt();

        System.out.print("Enter 1st Number = ");
        int num2 = sc.nextInt();

        int result = sumOfTwoNumbers(num1, num2);
        System.out.println("The sum of " + num1 + " and " + num2 + " is = " + result);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>java sumOfTwoNumbers
Enter 1st Number = 35
Enter 1st Number = 40
The sum of 35 and 40 is = 75
```

QUESTION – 2 - Simple Age Checker (Using a Method)

```
import java.util.Scanner;

class checkAgeUsingTernary
{
    void checkAgeCategory(int age) {

        String category = (age < 18)? "You are a Minor" : (age > 60)? "You are a Senior Citizen" : "You are an Adult";

        System.out.println(category);
    }

    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your Age: ");
        int age = sc.nextInt();

        checkAgeUsingTernary obj = new checkAgeUsingTernary();
        obj.checkAgeCategory(age);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>javac checkAgeUsingTernary.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>java checkAgeUsingTernary
Enter your Age: 16
You are a Minor

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>javac checkAgeUsingTernary.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>java checkAgeUsingTernary
Enter your Age: 47
You are an Adult

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>javac checkAgeUsingTernary.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>java checkAgeUsingTernary
Enter your Age: 68
You are a Senior Citizen
```

QUESTION – 3 - Print Even Numbers (Using while Loop)

```
class evenNumbers
{
    void printEvenNumbers() {
        int num = 1;
        while (num <= 50) {

            if(num % 2 == 0){
                System.out.println(num);
            }
            num++;
        }
    }

    public static void main(String[] args) {
        evenNumbers obj = new evenNumbers();
        obj.printEvenNumbers();
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>javac evenNumbers.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>java evenNumbers
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
```

QUESTION – 4 - User Input for Positive Numbers (Using do-while Loop)

```
import java.util.Scanner;

class positiveNumber
{
    static int askForPositiveNumber(Scanner sc) {
        int num;

        do {
            System.out.print("Enter a Positive Number: ");
            num=sc.nextInt();
        }while (num <= 0);

        return num;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int result = askForPositiveNumber(sc);

        System.out.println("You entered: " + result);
        sc.close();
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>javac positiveNumber.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>java positiveNumber
Enter a Positive Number: -78
Enter a Positive Number: 0
Enter a Positive Number: -3
Enter a Positive Number: 72
You entered: 72
```

QUESTION – 5 - Print Multiplication Table (Using for Loop)

```
import java.util.Scanner;

class multiplicationTable
{
    void printMultiplicationTable(Scanner sc) {

        System.out.print("Enter a Number: ");
        int num = sc.nextInt();

        for(int i=1; i<=10; i++) {
            System.out.println(num + " * " + i + " = " + (num*i));
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        multiplicationTable obj = new multiplicationTable();
        obj.printMultiplicationTable(sc);
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>javac multiplicationTable.java
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>java multiplicationTable
Enter a Number: 12
12 * 1 = 12
12 * 2 = 24
12 * 3 = 36
12 * 4 = 48
12 * 5 = 60
12 * 6 = 72
12 * 7 = 84
12 * 8 = 96
12 * 9 = 108
12 * 10 = 120
```

QUESTION – 6 - Calculate the Sum of Numbers from 1 to N (Using for Loop)

```
import java.util.Scanner;

class sumOfnNumbers
{
    static int calculateSum(Scanner sc) {

        System.out.print("Enter a Number: ");
        int num = sc.nextInt();

        int sum = 0;
        for(int i=1; i<=num; i++) {
            sum = sum + i;
        }
        return sum;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int result = calculateSum(sc);

        System.out.println("Sum = " + result);
        sc.close();
    }
}
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>javac sumOfnNumbers.java
```

```
C:\Users\Anjali\OneDrive\Desktop\CDAC Coding>java sumOfnNumbers
```

```
Enter a Number: 12
```

```
Sum = 78
```

BONUS PROBLEM: MENU-DRIVEN JAVA PROGRAM (SWITCH CASE)

```
import java.util.Scanner;

class menuDrivenProgram {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int choice;

        do {

            System.out.println("Enter you Choice!");

            System.out.println("1: Grade Evaluation System ");

            System.out.println("2: Leap Year Check ");

            System.out.println("3: Day of the week ");

            System.out.println("4: Identify Default Values of Variables");

            System.out.println("5: Exit! ");

            choice = sc.nextInt();

            switch(choice){

                case 1:

                    System.out.print("Enter your marks in Maths: ");

                    double marks1 = sc.nextInt();

                    System.out.print("Enter your marks in English: ");

                    double marks2 = sc.nextInt();

                    System.out.print("Enter your marks in Science: ");

                    double marks3 = sc.nextInt();

                    double average = (marks1 + marks2 + marks3)/3;
```

```
System.out.println("Average marks: " + average);
```

```
if (average >= 90) {
```

```
    System.out.println("GRADE: A");
```

```
}
```

```
else if (average >= 70) {
```

```
    System.out.println("GRADE: B");
```

```
}
```

```
else if (average >= 50) {
```

```
    System.out.println("GRADE: C");
```

```
}
```

```
else if (average >= 30) {
```

```
    System.out.println("GRADE: D");
```

```
}
```

```
else {
```

```
    System.out.println("FAIL");
```

```
}
```

```
break;
```

case 2:

```
System.out.print("Enter a year: ");
```

```
int year = sc.nextInt();
```

```
if(year % 4 == 0 && year % 100 != 0 || year % 400 == 0) {
```

```
    System.out.println(year + " is a Leap Year");
```

```
}
```

```
else{
```

```
    System.out.println(year + " is not a Leap Year");
```



```
}  
break;
```

case 3:

```
int day;  
do{  
    System.out.print("Enter a day[1 to 7]: ");  
    day = sc.nextInt();
```

```
}while(day < 1 || day > 7);
```

```
switch(day){  
    case 1:  
        System.out.println("The day is Monday");  
        break;  
    case 2:  
        System.out.println("The day is Tuesday");  
        break;  
    case 3:  
        System.out.println("The day is Wednesday");  
        break;  
    case 4:  
        System.out.println("The day is Thursday");  
        break;  
    case 5:  
        System.out.println("The day is Friday");  
        break;  
    case 6:  
        System.out.println("The day is Saturday");  
        break;  
    case 7:  
        System.out.println("The day is Sunday");
```

```

        break;

    default:

        System.out.println("Invalid Choice. Please enter days between 1 to 7.");

    }

    break;

    case 4:

        System.out.println("Default values of instance variables in Java");

        defaultValues obj = new defaultValues();

        obj.display();

        break;

    case 5:

        System.out.println("Exiting the Program");

        break;

    default:

        System.out.println("Invalid Choice");

    }

}while(choice != 5);

}

}

```

```

class defaultValues {

    byte a;

    short b;

    int c;

    long d;

    float e;

    double f;

    char g;

    boolean h;

```

```

void display() {

    System.out.println("byte   : " + a);
    System.out.println("short  : " + b);
    System.out.println("int    : " + c);
    System.out.println("long   : " + d);
    System.out.println("float  : " + e);
    System.out.println("double : " + f);
    System.out.println("char   : " + (int) g);
    System.out.println("boolean : " + h);
}
}

```

```

C:\Users\Anjali\OneDrive\Desktop\CDAC Coding Assignments>java menuDrivenProgram
Enter you Choice!
1: Grade Evaluation System
2: Leap Year Check
3: Day of the week
4: Identify Default Values of Variables
5: Exit!
4
Default values of instance variables in Java
byte      : 0
short     : 0
int       : 0
long      : 0
float     : 0.0
double    : 0.0
char      : 0
boolean   : false
Enter you Choice!
1: Grade Evaluation System
2: Leap Year Check
3: Day of the week
4: Identify Default Values of Variables
5: Exit!
3
Enter a day[1 to 7]: 0
Enter a day[1 to 7]: 9
Enter a day[1 to 7]: 4
The day is Thursday
Enter you Choice!
1: Grade Evaluation System
2: Leap Year Check
3: Day of the week

```